Akash Sharma

Contact

akashsharma@cmu.edu (last updated May 2025)
`https://akashsharma02.github.io`

Education

Carnegie Mellon University

Ph.D. student in Robotics, Advisor: Michael Kaess — 2021 - 2026
Proposed thesis: `Self-supervised perception for tactile dexterity`
Thesis Committee: `Michael Kaess`
`Shubam Tulsiani`
`Guanya Shi`
`Mustafa Mukadam (Amazon)`
`Jitendra Malik (UC Berkeley, Meta)`
GPA: `4.11/4.33`

M.S. in Robotics, Advisor: Michael Kaess — 2019 - 2021
Thesis: `Incorporating semantic structure in SLAM`

Sri Jayachamarajendra College of Engineering

B.E. in Electronics and Communication — 2013 - 2017
GPA: `9.61/10.00`

Research Experience

FAIR at Meta, Pittsburgh, PA
Visiting Researcher — 2023 - 2025
Perception for dexterous manipulation: Working on self-supervised (SSL) representation learning for tactile sensors. Currently working on generative modeling of dexterous manipulation tasks using vision and touch modalities. Published at CoRL 2024.

The Robotics Institute, CMU, Pittsburgh, PA
Graduate Student Researcher — 2019 - 2025
Semantic SLAM with Object Landmarks: Worked on a SLAM system that reconstructs an environment as a collection of objects. The system fuses sensor data from RGBD cameras, object detection and segmentation networks in a non-linear optimization framework to estimate object shape and color, 6DoF pose and camera poses. Published at ICRA 2021

Publications

[In submission] Jay Karhade, **Akash Sharma**, Y. Zhang, N. V. Keetha“Any4D: A Unified Model for 4D Scene Reconstruction with Flexible Conditioning on Scene and Camera Priors.”

[In submission] Carolina Higuera*, **Akash Sharma***, , T. Fan*, C. K. Bodduluri, B. Boots, M. Kaess, M. Lambeta, T. Wu, M. Mukadam, Z. Liu, F. R. Hogan.“Tactile Beyond Pixels: Multisensory Touch Representations for Robot Manipulation.” (* equal contribution)

[In submission] **Akash Sharma**, C. Higuera, C. K. Bodduluri, T. Fan, T. Hellebrekers, M. Lambeta, B. Boots, M. Kaess, M. Kalakrishnan, T. Wu, M. Mukadam. “Sparsh-skin: Perception via Self-supervision for Dexterous hands covered with tactile skin”

Zhao-Heng Yin, C. Wang, L. Pineda, F. Hogan, C. K. Bodduluri, **Akash Sharma**, P. Lancaster, I. Prasad, M. Kalakrishnan, J. Malik, M. Lambeta, T. Wu, P. Abbeel, M. Mukadam. “DexterityGen: Foundation Controller for Unprecedented Dexterity” *Robotics: Science and Systems (RSS), 2025* | **pdf**

Carolina Higuera*, **Akash Sharma***, C. K. Bodduluri, T. Fan, M. Kalakrishnan, M. Kaess, B. Boots, M. Lambeta, T. Wu, M. Mukadam. “Sparsh: Self-supervised touch representations for vision-based tactile sensing.” *8th Annual Conference on Robot Learning (CoRL), 2024* | **pdf** | **code** (* equal contribution)

Ming-Fang Chang, **Akash Sharma**, Michael Kaess, Simon Lucey. “Neural Radiance Fields with LiDAR Maps” *IEEE/CVF International Conference on Computer Vision (ICCV), 2023* | **pdf**

Ruoyang Xu, Wei Dong, **Akash Sharma**, Michael Kaess. “Learned Depth Estimation of 3D Image Radar for Indoor Mapping” *IEEE/RSJ Intl. Conf. on Intelligent Robots and Systems (IROS) 2022* | **pdf**

Akash Sharma, Wei Dong, Michael Kaess. “Compositional Scalable Object SLAM”*IEEE Intl. Conf. on Robotics and Automation (ICRA) 2021* | **pdf** | **code**

Professional Experience

Reality Labs Research, Meta, Redmond, WA
Research Scientist Intern, Surreal Vision team — Summer 2022
Representation Learning for robust odometry: Proposed an end-to-end transformer that learns a 3D representation from a stream of multi-modal data (vision and IMU) to predict odometry. Predicted odometry was auto-regressively composed to estimate the trajectory of (`Project Aria`) AR glasses.

Fyusion Inc., San Francisco, CA
Research Intern — Summer 2021
Free viewpoint view synthesis for car interiors: Developed a neural radiance field representation-based novel view synthesis method tuned for free viewpoint synthesis specific for 360° outward facing cameras. I experimented with multiple different methods in both image-based rendering as well as physically based rendering.

OpenCV (Google Summer of Code), Virtual / Pittsburgh, PA
Student Developer — Summer 2020
3D Spatial Hashing for Large scale dense reconstruction: Implemented and extended Kinect Fusion using spatial hashing and submap based mapping for reconstruction of large scale environments.(`blog`)

Infinera, Bangalore, India
Software Engineer — 2017 - 2019
Improved the optical device infrastructure: Developed a configurable system infrastructure software for optical amplifier devices to monitor faults and performance. *Enabled fast optical traffic startup*: Bypassed an auto-discovery mechanism in the optical amplifier hardware for improved laser power control and faster optical power startup. Mentored incoming undergraduate students in the optical line system team.

Press Coverage

Advancing embodied AI through progress in touch perception, dexterity, and human-robot interaction

AI at Meta, blog	2024
TechRadar	2024
VentureBeat	2024
Business Today	2024
Interesting Engineering	2024
Maginative	2024
The Decoder	2024
MarkTechPost	2024

Teaching

Teaching Assistant: Probabilistic Graphical Models (Prof. Andrej Risteski) — Fall 2022
Teaching Assistant: Geometry-based methods for Computer Vision (Prof. Michael Kaess) — Fall 2021
Teaching Assistant: Robot Localization and Mapping (Prof. Michael Kaess) — Fall 2020
"*Guest lecture on algorithms for dense SLAM*"
- 16833 - Robot Localization and Mapping, CMU — 2022
- 16833 - Robot Localization and Mapping, CMU — 2020

Talks

"*Self-supervised perception for tactile dexterity*".
- Neuroagent lab, CMU — 2025

"*Sparsh: SSL touch representations for tactile sensing*".
- Franka Robotics, GmbH — 2024
- Conference, FAIR at Meta — 2024
- FAIR Embodied AI seminar — 2024

"*Self-supervised learning in Vision*". GUM Reading Group, Meta. — 2024
"*ViTs for mean-teacher distillation with no labels*". Misc-Reading Group at CMU. — 2022
"*Learning a multimodal state representation for odometry estimation*". Surreal team, Meta. — 2022

Service

Thesis committee: Vivek Roy (Now @ Apple) — 2022
Admissions committee: MS Robotics — 2021 - 2022
Conference reviewer: CVPR 24-25, NeurIPS 24, RA-L 23-24, IROS 2022, ICRA 22 - 21

Mentorship

Angela Chen, RI (PhD) Peer Mentor Program — 2022
Mrinal Verghese, RI (MSR) Peer Mentor Program — 2020
Mary Hatfalvi, RI (MSR) Mentoring Program — 2020